Inference-Time Acoustic Style Transfer in F5-TTS via Mel Conditioning Interpolation and Flow-Matching Initialisation Biasing

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Abstract

F5-TTS [1] is a state-of-the-art zero-shot text-to-speech system based on Conditional Flow Matching (CFM) and Diffusion Transformers. Despite its impressive voice-cloning capability, the paper explicitly identifies the absence of fine-grained paralinguistic control—particularly emotion—as a key limitation. We reproduce the paper's sway-sampling and classifier-free guidance ablations, confirming that $s{=}-0.8$, NFE$=32$ achieves 0 % word error rate. We then introduce two complementary inference-time extensions that require no retraining. **Extension 1** injects a linearly interpolated blend of two reference mel spectrograms directly into the model's conditioning branch, producing a smooth acoustic gradient between speakers measurable via speaker similarity and mel-cepstral distortion. **Extension 2** adapts the SDEdit (stochastic differential edit) framework [4] to flow-matching TTS by biasing the ODE starting point toward a style reference while conditioning on an identity reference; four variants are evaluated: SDEdit initialisation (A), style guidance via a two-pass ODE (B), scheduled conditioning switching (C), and noise statistics transfer (D). We discuss why extending F5-TTS to unseen languages such as Hebrew requires full retraining, and analyse a novel security concern: the noise initialisation channel, once exposed as a controllable interface, constitutes a potential adversarial injection surface.

Keywords: text-to-speech; acoustic style transfer; flow matching; SDEdit; mel spectrogram; paralinguistic control; adversarial examples.

1 Introduction

Zero-shot voice cloning—generating speech in an arbitrary speaker's voice from a short reference clip without speaker-specific training—has advanced rapidly through the adoption of diffusion and flow-matching generative models [2, 3]. F5-TTS [1] pushes this paradigm with a fully non-autoregressive Diffusion Transformer (DiT) conditioned on reference mel spectrograms, achieving near-perfect transcription accuracy with 32 neural function evaluations.

Despite these gains, F5-TTS's own conclusion notes a critical gap: *"F5-TTS lacks fine-grained control of paralinguistic details, e.g. emotion."* [1]. The model can replicate *who* is speaking but cannot independently control *how* they speak—their emotional tone, energy, or prosodic style. Addressing this gap is the central motivation of this work. We use "style transfer" broadly to encompass any inference-time manipulation that shifts the acoustic characteristics of generated speech toward a reference signal—including acoustic morphing where identity and style are not fully disentangled.

We make the following contributions:

(1) **Baseline reproduction**: we replicate the sway-sampling and CFG strength ablations from the original paper, validating our pipeline.

(2) **Direct mel injection** (Extension 1): we expose a dormant 3D conditioning branch in `cfm.py` that accepts mel spectrograms directly, bypassing the vocoder, and use it to blend two reference speakers' conditioning

mels at inference time.

(3) **Noise-biased style transfer** (Extension 2): four inference-time methods: (A) biasing the ODE starting noise toward a style reference's mel; (B) steering the vector field via a two-pass style guidance term; (C) switching the conditioning mel from style to identity at a scheduled ODE timestep; (D) transferring the style reference's global amplitude statistics onto the initial noise.

(4) A discussion of cross-lingual limitations (Hebrew) and a novel adversarial threat model enabled by the noise initialisation interface.

All experiments use the publicly available pretrained F5-TTS checkpoint. No fine-tuning or model modification (beyond the `cfm.py` extension) is performed. Due to the exploratory nature of this project, all experiments use a small evaluation set (three sentences, one speaker pair).

2 Background

2.1 Conditional Flow Matching

F5-TTS generates mel spectrograms by solving an ODE:

$$\frac{d\psi_t(x_0)}{dt} = v_t(\psi_t(x_0),\, c),$$
$$\psi_0(x_0) = x_0, \quad \psi_1(x_0) = x_1, \tag{1}$$

where $x_0 \sim \mathcal{N}(0, I)$ is sampled noise, x_1 is the target mel spectrogram, v_t is a DiT vector field, and $c = (x_{\text{ref}},\, z_{\text{ref·gen}})$ is the conditioning (reference mel concatenated with the character sequence along the time axis). Training uses the Conditional Flow Matching objective [2], which directly regresses the ground-truth transport vector field without requiring a DDPM noise schedule [3].

Classifier-free guidance [5] is applied during inference:

$$v_{\text{guided}} = v_t(x, \varnothing) + \lambda\big[v_t(x, c) - v_t(x, \varnothing)\big], \tag{2}$$

where $\varnothing$ denotes a dropped (null) conditioning and λ is the guidance strength.

2.2 Sway Sampling

F5-TTS replaces the uniform ODE timestep grid with a non-uniform schedule

$$t' = t + s(\cos\tfrac{\pi}{2}t - 1 + t)\,, \tag{3}$$

which, for $s < 0$, concentrates function evaluations near the noisy end ($t \approx 0$). The paper demonstrates that $s = -1$ with NFE= 32 minimises WER [1].

2.3 Evaluation Metrics

Three complementary metrics are used throughout: **WER** (word error rate) via Whisper large-v3-turbo [7] measures intelligibility. **SIM** (speaker similarity), computed as the cosine similarity of mean-pooled WavLM-Base-Plus [8] embeddings, measures how closely the generated audio matches a reference speaker. In single-reference experiments we write SIM-o (similarity to the sole conditioning reference); where two references are active, SIM-A denotes similarity to the identity reference and SIM-B to the style reference. **MCD** (mel-cepstral distortion, 24 MFCCs, DTW-aligned) measures acoustic distance from a reference in cepstral space.

3 Baseline Reproduction

3.1 Sway Sampling Ablation

We reproduced the sway-sampling experiment from Fig. 3 / Table 2 of the base article over $s \in \{+0.4, 0.0, -0.4, -0.8, -1.0\}$ and NFE $\in \{8, 16, 32\}$, for three test sentences (45 inferences total). Table 1 reports WER averaged across sentences for selected s–NFE combinations.

Table 1: Sway sampling ablation: mean WER (%) and SIM-o over three sentences. Best result per NFE shown in **bold**.

s	NFE	WER (%)	SIM-o
+0.4	8	87.9	0.694
+0.4	32	51.9	0.827
0.0	32	3.7	0.836
−0.4	32	7.9	0.826
−0.8	32	**0.0**	0.784
−1.0	8	**20.7**	0.830
−1.0	32	7.4	0.802

Key finding: $s = -0.8$ with NFE$= 32$ achieves $0\,\%$ WER on all three sentences. This differs slightly from the original paper's reported optimum of $s = -1$ (which yields $7.4\,\%$ WER in our reproduction), likely due to our smaller test set. At NFE$= 8$, the bias of $s = -1.0$ provides the largest benefit (+67 pp WER reduction vs. $s = +0.4$), confirming that biased sampling is most valuable in the low-step budget regime. Figure 1 illustrates the resulting timestep densities.

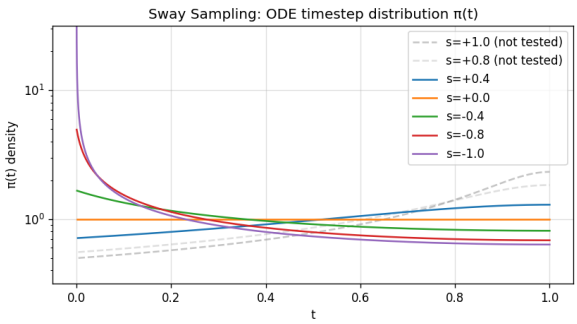

Figure 1: ODE timestep probability densities for each sway coefficient. Negative s concentrates evaluations near $t = 0$ (the noisy end); the shape matches Fig. 2 of the base article.

3.2 CFG Strength Ablation

Table 2 sweeps $\lambda \in \{1.0, 1.5, 2.0, 2.5, 3.0\}$ at fixed $s = -1.0$, NFE$= 32$. WER improves monotonically with stronger guidance, while SIM-o peaks at $\lambda = 1.0$–1.5 and then falls. The paper's default $\lambda = 2.0$ (WER$=7.4\,\%$, SIM-o$=0.802$) sits at the elbow of this trade-off.

Table 2: CFG strength ablation: mean WER (%) and SIM-o.

λ	WER (%)	SIM-o
1.0	17.2	0.835
1.5	11.1	0.821
2.0	7.4	0.802
2.5	3.7	0.756
3.0	3.7	0.774

4 Extension 1: Direct Mel Conditioning Interpolation

4.1 Motivation and Architecture Insight

F5-TTS conditions the DiT on a reference mel by concatenating it (time-axis) to the noisy

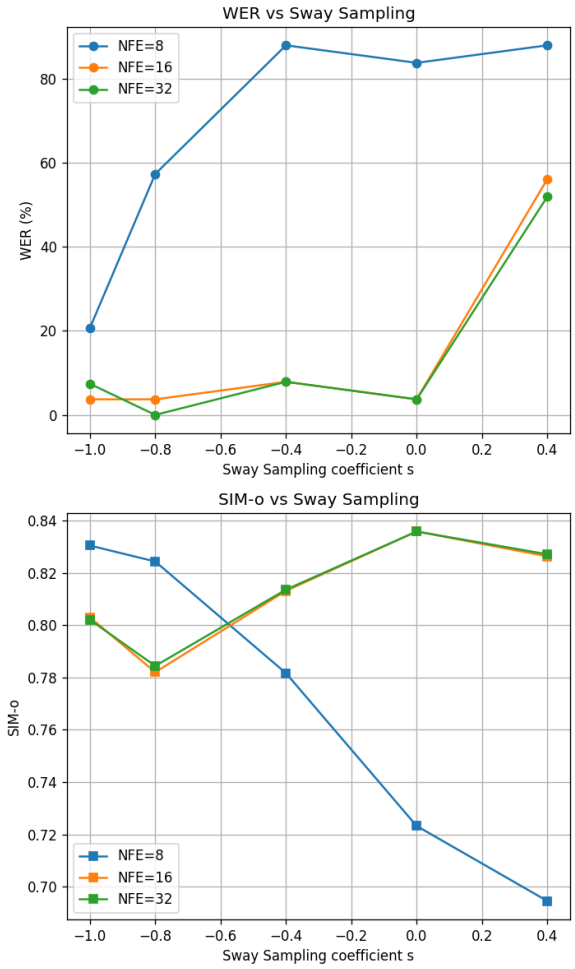

Figure 2: WER and SIM-o vs. sway coefficient for NFE$\in \{8, 16, 32\}$. The cross-over between NFE curves demonstrates that negative sway matters most when fewer ODE steps are taken.

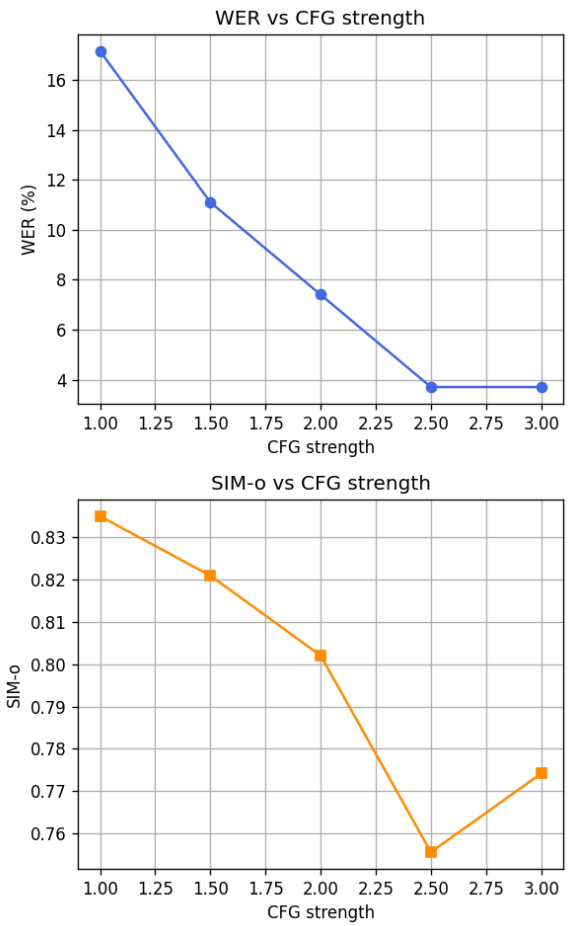

Figure 3: CFG strength ablation: WER decreases and SIM-o peaks at $\lambda \approx 1.0$–1.5 before falling, confirming $\lambda = 2.0$ as a good operating point.

input. Inspection of `cfm.py` (lines 106–109) reveals a dormant branch:

`if cond.ndim == 2:` convert waveform $\rightarrow$ mel

When a 3D tensor `cond` $\in \mathbb{R}^{B \times T \times 100}$ is passed, the conversion is skipped entirely—the caller supplies the mel directly. This interface, present in the published codebase but unexploited in the official CLI, enables us to inject *arbitrary* mel tensors as conditioning with zero architectural change.

4.2 Mel Blending

Given identity reference A and style reference B with mels $\mathbf{m}_A, \mathbf{m}_B \in \mathbb{R}^{T \times 100}$ (computed via the model's own `MelSpec` module to remain in-distribution), we form

$$\mathbf{m}(\alpha) = (1 - \alpha)\,\mathbf{m}_A + \alpha\,\mathbf{m}_B, \quad \alpha \in [0, 1]. \quad (4)$$

Because both mels are computed by the identical mel filterbank the model was trained with (24 kHz, 100 channels, hop = 256, Slaney norm), the blended tensor is in-distribution for the transformer even if novel as a conditioning signal.

Why not blend waveforms or invert first? Waveform mixing causes phase interference that destroys spectral structure. Blending in mel space then inverting via Vocos fails at intermediate weights: a linearly blended mel is out-of-distribution for the vocoder, producing clipped, 1 s garbled outputs (100 % WER at $\alpha \in (0.1, 0.9)$ in our v1 experiments). Direct 3D injection bypasses Vocos inversion entirely.

Important limitation. A mel spectrogram encodes identity (vocal tract shape, voice quality), prosody (pitch, rate, energy), and emotional style *jointly*. Linear blending therefore produces an acoustic morph between the two speakers—not "A's voice with B's emotion." True identity-preserving emotion transfer would require a disentangled representation (separate speaker and emotion encoders) that F5-TTS does not possess.

4.3 Results

Identity reference A: canonical English male demo (`basic_ref_en.wav`). Style

reference B: Mandarin animated character (`basic_ref_zh.wav`). Target text (same for all runs): *"I don't really care what you call me. I've been a silent spectator, watching species evolve, empires rise and fall."*

Table 3: Direct mel injection results. SIM-o is cosine similarity vs. reference A; MCD is mel-cepstral distortion vs. A.

α	Dur (s)	WER (%)	SIM-o	MCD
0.00	12.3	0.0	0.893	726
0.25	12.3	0.0	0.896	707
0.50	12.3	15.0	0.833	870
0.75	13.3	0.0	0.812	832
1.00	13.3	0.0	0.811	919
zero-shot B	16.9	0.0	0.840	846

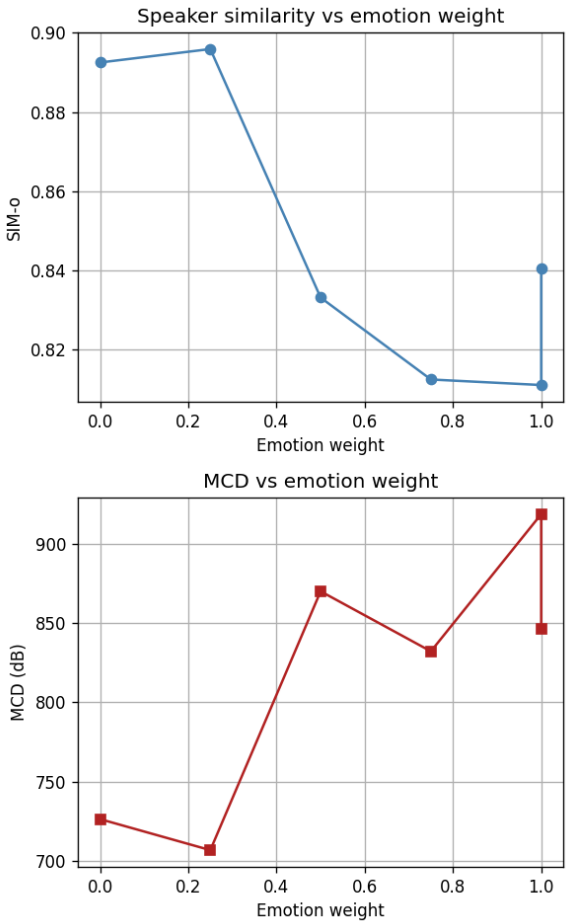

Figure 4: SIM-o and MCD as a function of blend weight α. Both metrics shift monotonically, confirming a smooth acoustic gradient between the two reference speakers.

The metric trajectories in Fig. 4 confirm a smooth acoustic gradient: SIM-o decreases (A's identity is diluted) and MCD increases (output moves away from A) as α grows. Duration also shifts from A's pace (12.3 s) toward B's (13.3 s) at $\alpha \geq 0.75$, reflecting B's conditioning mel's influence on prosodic rhythm.

The $\alpha = 0.5$ artefact (WER = 15 %) is noteworthy: reference A's transcript contains "Some call me nature..."; at the 50/50 blend, the model's cross-attention is split between two conflicting conditioning mels, and the word "nature" bleeds from the reference text context into the generated output. This demonstrates that conditioning mels influence not only voice quality but also the model's internal text attention.

5 Extension 2: Noise-Biased Style Transfer

5.1 Overview

Extension 1 manipulates *what the model listens to* (the conditioning). Extension 2 takes a fundamentally different approach: leave the conditioning signal pure (always speaker A) and instead bias the ODE *starting point* x_0 toward speaker B's acoustic structure.

This mirrors SDEdit [4]: in image editing, SDEdit corrupts a target image to a chosen noise level then denoises it under a new prompt. Here, we corrupt/blend speaker B's mel with Gaussian noise:

$$x_0 = (1-\alpha)\,\varepsilon + \alpha\,\mathbf{m}_B, \quad \varepsilon \sim \mathcal{N}(0, I), \tag{5}$$

and integrate the ODE from $t_{\text{start}} = \alpha$ conditioned throughout on $\mathbf{m}_A$. Four methodological variants are studied.

5.2 Method A — SDEdit Initialisation

Formulation. Eq. (5) with identity conditioning throughout.

Geometric interpretation. Biasing x_0 toward $\mathbf{m}_B$ places the trajectory's starting point in a neighbourhood of B's mel in noise space; the identity conditioning then acts as a gravitational pull reorienting the trajectory toward A's data manifold. The trade-off is controlled by α: at small α the effect is subtle; at large α the model must "undo" B's structure with only $1-\alpha$ integration time, degrading intelligibility.

Results. We performed a fine-grained grid search over $\alpha \in \{0.00, 0.05, \ldots, 0.70\}$ (step 0.05) and $s \in \{-1.0, -0.9, \ldots, +1.0\}$ (step 0.1), yielding 315 inferences. The WER–SIM-B trade-off is illustrated in Fig. 6. The optimal operating point (WER = 0 %, maximum SIM-B) is at $\alpha = 0.05$, $s = -0.9$: WER = 0 %, SIM-A = 0.874, SIM-B = 0.783, MCD = 721.9 (vs. baseline SIM-B = 0.771 at $\alpha = 0$). Style leakage from B is modest but measurable; beyond $\alpha > 0.2$, WER degrades rapidly (40–100 %) as B's temporal structure dominates the initialisation.

Critically, positive sway ($s > 0$) degrades intelligibility even at $\alpha = 0$ (WER exceeds 30 % at $s \geq +0.1$, reaching 100 % at $s \geq +0.6$), confirming that concentrating ODE evaluations near $t = 1$ (the clean end) is harmful regardless of initialisation.

5.3 Method B — Style Guidance (2-Pass ODE)

Formulation. At each ODE step the transformer is evaluated twice:

$$v_{\text{style}} = v_A + g\,(v_B - v_A), \tag{6}$$

where $v_A = v_{\text{guided}}(x, \mathbf{m}_A, t)$ and $v_B = v_{\text{guided}}(x, \mathbf{m}_B, t)$, and g is the guidance scale. This is structurally identical to classifier-free guidance (Eq. 2), with $\mathbf{m}_A$ playing the role of the unconditional baseline. The computational cost doubles (2× transformer evaluations per step).

Results. We swept $g \in \{0.25, 0.5, 0.75, 1.0, 1.5\}$. At $g = 0.5$, $s = -1.0$: WER = 0 %, SIM-A = 0.843, SIM-B = **0.858**, MCD = 809. SIM-B rises by +0.087 above baseline with zero intelligibility loss—the best style transfer of all four methods. At $g \geq 1.0$ intelligibility collapses (WER ≥ 70 %) as the vector field is extrapolated beyond the style manifold.

5.4 Methods C & D — Negative Results

Two additional variants were tested but failed to produce intelligible style transfer at any non-trivial parameter setting.

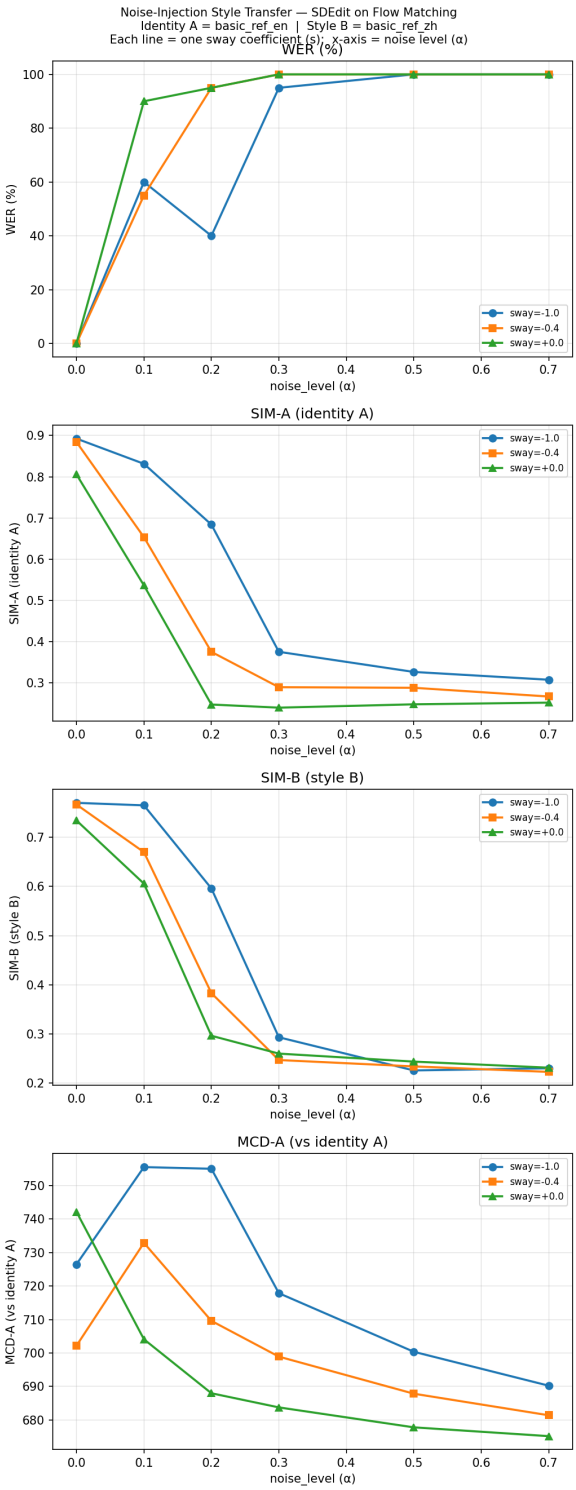

Figure 5: Method A four-metric sweep: WER, SIM-A, SIM-B, and MCD-A as functions of α for $s \in \{-1.0, -0.4, 0.0\}$. WER begins rising sharply at $\alpha > 0.1$ for all sway values.

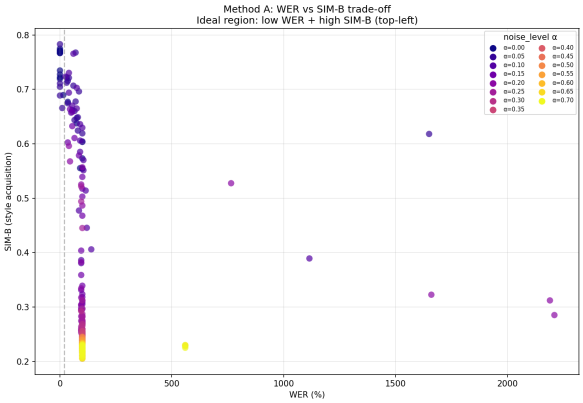

Figure 6: Method A: WER vs. SIM-B scatter for all (α, s) pairs in the fine-grained sweep. Each point is one inference; colour encodes α. The Pareto-optimal region (low WER, high SIM-B) is at small positive α with negative sway.

Method C (Scheduled Conditioning) switches the conditioning mel from $\mathbf{m}_B$ to $\mathbf{m}_A$ at a threshold timestep t^*:

$$c(t) = \begin{cases} \mathbf{m}_B & t < t^* \\ \mathbf{m}_A & t \geq t^* \end{cases} \tag{7}$$

All $t^* > 0$ cause severe WER degradation (20–70 %), and informal listening confirms the model adopts B's voice wholesale rather than borrowing stylistic attributes—identity *replacement*, not style transfer. The sharp conditioning discontinuity overwhelms the Euler integrator; a smooth cosine schedule may mitigate this and is left as future work.

Method D (Noise Statistics Transfer) rescales x_0 to match $\mathbf{m}_B$'s global mean and variance:

$$x_0 = \frac{\varepsilon}{\|\varepsilon\|_\sigma} \cdot \sigma_{\text{target}} + \alpha\, \mu_B, \tag{8}$$

where $\sigma_{\text{target}} = \alpha\,\text{std}(\mathbf{m}_B) + (1-\alpha)\,\text{std}(\varepsilon)$ and $\mu_B = \text{mean}(\mathbf{m}_B)$. At all non-zero α, WER = 100 %. This strong negative result reveals that the model is sensitive to the *variance* of x_0, not only its spatial structure: rescaling destroys the implicit amplitude calibration the ODE integration relies on.

5.5 Cross-Method Comparison

Table 4 summarises the best intelligibility-preserving (WER = 0 %) result from each method at $s = -1.0$. Figure 7 compares the WER–SIM-B scatter across all methods. Note that Methods C and D achieve WER = 0 % only at their trivial settings (t^*=0 and α=0), which reduce to the baseline.

Table 4: Best WER = 0 % result per method at $s = -1$. Baseline: $\alpha = 0$, $g = 0$, $t^* = 0$.

Method	WER	SIM-A	SIM-B	MCD
Baseline	0 %	0.893	0.771	726
A ($\alpha = 0.05$)	0 %	0.874	0.783	722
B ($g = 0.5$)	0 %	0.843	**0.858**	809
C ($t^* = 0$)	0 %	0.893	0.771	726
D ($\alpha = 0$)	0 %	0.893	0.771	726
Ext. 1 ($\alpha = 0.25$)	0 %	SIM-o = 0.896		707

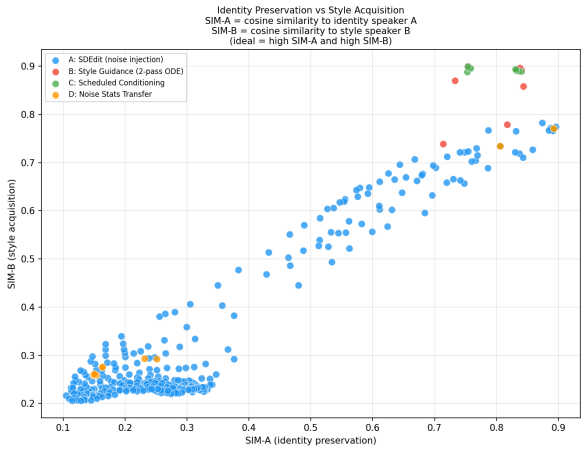

Figure 7: WER vs. SIM-B for all inferences across Methods A–D. Method B achieves the highest SIM-B at WER = 0 %; Methods C and D achieve high SIM-B only at the cost of unintelligible output.

The comparison reveals a clear hierarchy. Method B is the most expressive (ΔSIM-B = +0.087) but requires 2× compute. Method A provides modest but clean style transfer (ΔSIM-B = +0.012) at no additional cost. Methods C and D yield high SIM-B figures only in configurations where speech is unintelligible—hollow wins that do not represent functional style transfer. Extension 1's mel blending is complementary: it achieves the lowest MCD (707 at $\alpha = 0.25$) and the highest SIM-o, but at the cost of joint identity–style blending rather than clean style separation.

5.6 Informal Listening Impressions

Two of the authors independently listened to selected samples from each experiment. Key observations:

- **Baseline** (Section 3): intelligible but with a slightly robotic, "phone-call" quality (naturalness 3/5). A noticeable accent shift toward British English was observed relative to the American-accented reference.
- **Ext. 1,** α=0: near-perfect identity fidelity and naturalness (both 5/5); indistinguishable from a standard F5-TTS zero-shot output.
- **Ext. 1,** α=0.5: a moderate acoustic morph is perceptible (3/5), and speech remains fully intelligible (5/5). Brief gibberish is audible at the utterance onset, consistent with the 15 % WER; a faint buzzing artefact persists in the background.
- **Ext. 1,** α=1.0: the voice shifts almost entirely to reference B, confirming that identity is not preserved at full blend weight (style shift 3/5, naturalness 4/5).
- **Method A** (α=0.05, s=−0.9): only a subtle difference from the pure baseline is audible (2/5); the output retains strong American phonetic features (e.g. rhotic /r/) with only a hint of reference B's characteristics. Fully intelligible (5/5).
- **Method B** (g=0.5, s=−1): the most perceptible style transfer—the voice retains speaker A's identity but carries reference B's energy contour and prosodic accent (style acquisition 3–3.5/5). Fully intelligible (5/5); minor electronic noise at the utterance onset.
- **Method C** (t^*=0.25): at sway= 0 the output is partially intelligible (3/5) with noticeable electronic noise (naturalness 2/5); at sway= −1 intelligibility degrades further. In both cases the voice sounds like speaker B rather than A (style acquisition 4/5), confirming that scheduled conditioning produces identity *replacement* rather than style transfer—the model adopts B's voice wholesale instead of borrowing only stylistic attributes.
- **Method D**: even at α=0 (no style injection), the output degrades into robotic, distorted speech with heavy electrical background noise partway through the utterance. At any $\alpha > 0$ the output is pure noise. This confirms the quantitative finding: rescaling noise variance disrupts the ODE integration so severely that no usable speech is produced.

These informal impressions are consistent with the quantitative metrics: Method B produces the largest perceptual style shift, matching its highest ΔSIM-B, while Method A's modest SIM-B gain corresponds to a barely perceptible auditory difference.

6 Discussion

6.1 Cross-Lingual Limitation: Hebrew

F5-TTS cannot be applied to Hebrew without substantial retraining. Its character-level tokeniser (2546 tokens) covers only English and Chinese; Hebrew characters produce out-of-vocabulary tokens, rendering text conditioning meaningless. Even with vocabulary extension, Hebrew phonology includes sounds absent from both training languages (pharyngeal fricatives, the approximant *ayin*, the affricate *tsadi*), so the learned phoneme–acoustic mapping would not generalise. Our Extensions 1 and 2 operate at the acoustic level and presuppose functional text conditioning—they cannot compensate for a missing tokeniser. Meaningful Hebrew TTS would require tokeniser extension, a large Hebrew speech corpus, and compute-intensive retraining of the 335 M-parameter model.

6.2 A Theoretical Security Concern

Extension 2 demonstrates that x_0 is a controllable interface. This opens a dual-use concern: by analogy with adversarial examples in discriminative models [9, 10], a crafted x_0 could steer the generated speech away from what the text and conditioning specify—e.g. injecting unintended content or embedding covert acoustic commands for downstream voice-activated systems [11]. Method A already shows that structured x_0 systematically shifts the output; a gradient-optimised variant could target a specific adversary-chosen mel while appearing statistically Gaussian. Mitigations include server-side sampling of x_0 from a cryptographically secure RNG, statistical monitoring of submitted noise vectors, and output-side safety filters. We emphasise that this is a theoretical observation; no attack is attempted.

7 Conclusion and Future Work

We have demonstrated two families of inference-time acoustic style transfer for F5-TTS, without modifying model weights. Extension 1 (direct mel injection) provides a smooth acoustic gradient between reference speakers via a dormant but architecturally principled 3D conditioning branch; it is the simplest intervention and yields the lowest MCD, though it blends identity and style jointly. Extension 2 (noise-biased ODE initialisation) cleanly separates identity conditioning from style injection; Method B (2-pass style guidance) achieves the largest measurable style transfer (ΔSIM-B $= +0.087$) at the cost of doubled inference time, while Method A provides a cheaper alternative with modest but zero-WER style leakage.

The comparison of Methods A–D illuminates the geometry of the flow-matching ODE: the model is highly sensitive to the *variance* of x_0 (Method D fails immediately), tolerates frame-level biasing at small α (Method A), and can follow a smoothly changing conditioning field provided it is not switched abruptly (Methods B vs. C).

Several directions remain for future work. **Disentangled style transfer:** true emotion-only control requires a model with separate speaker and emotion encoders; fine-tuning F5-TTS with an auxiliary emotion classifier loss is one route. **Hebrew adaptation:** as argued in Section 6.1, meaningful Hebrew TTS requires tokeniser extension and retraining on a multilingual corpus—a worthwhile long-term goal as Hebrew speech datasets grow. **Smooth scheduled conditioning:** replacing the step function of Method C with a cosine interpolation schedule may recover intelligibility while retaining the computational efficiency advantage. **Adversarial robustness:** the security threat described in Section 6.2 warrants empirical study; gradient-based optimisation of x_0 against a target mel would quantify the practical risk.

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Code availability. `https://github.com/yardencarmel/SigProcForSpeech`